

DSA Software Engineering Event

Overview

- You will be assigned into groups of 6-8 people
- Develop a visualized DSA
- Select roles from PM, FE, BE, QA
- Write your personal documentation
- Perform a booth exhibition to showcase your program

Start and end time:

- Start: 2026/3/31
- End: 2026/5/19 (Exhibition Day!!!)

Full Spec: <https://hackmd.io/@Jenehsu0919/SJs5hGYcWI>

Group Formation & Pairing Up (Optional)

Groups will be assigned randomly by the TAs, but you have the option to pair up with **one** other student beforehand!

- Guaranteed Pairing: If you form a pair, you two will be guaranteed to be assigned to the same random group.
- Pairing up is **optional**.
- If you decide to pair up, only **one person** from the pair needs to fill out the form.
- Please fill out the form before 2026/4/5 23:59
- Groups will be announced on 2026/4/7

Pairing Form: <https://forms.gle/1CJFPhkJWUBQgSSs5>

Develop Visualized DSA

- Choose any Data Structure or Algorithm to implement
 - Doesn't need to be covered in class
- Visualize the DSA to make it easier to understand
 - Showcase the workflow through moving entities, color changes, animations, etc.
- Scratch is the default collaboration platform
 - You could choose other platform as long as you can achieve visualization and real-time graphic output
- Coolest DSA Voting will be hosted on the presentation date
 - Cool prizes and special snacks!!!

Role Selection

Roles to select:

- Project Manager (PM)
 - 1 person per group
- Frontend (FE)
 - 2-4 people per group
- Backend (BE)
 - 1-2 people per group
- Quality Assurance (QA)
 - 1-2 people per group

Project Manager (PM)

- Break down and distribute work segments to each members
- Define tasks and goals
- Set input and output constraints
- Be familiar with the responsibilities of FE, BE, and QA
- Schedule task and track process
- Make final decisions on issues or changes to the project plan
- Contact TAs via email if clarifications or assistance are needed
- Write an additional README document
- Submit a 30-Second Introduction Video

Frontend (FE)

- Develop the User Interface (UI) for user to interact with the application
- Make sure the movement, color, or any related visual feedback conveys how the DSA works
- Connect UI inputs to backend's DSA input with correctness
- FE does not handle DSA logic implementation

Backend (BE)

- Implement the DSA logic on the chosen platform
- Make sure the DSA works without bug given any legal operation
- BE does not handle UI design

Quality Assurance (QA)

- Test the program to make sure the program runs properly
- Understand the entire codebase to guarantee correctness and efficiency

Personal Document

- Personal Documentation (Everyone):
 - Include proof of your contributions to the project
 - Examples: Code snapshots, detected bugs, UI elements designed, group progress tracking, decision-making involvement, etc.
- Extra README (PM Only):
 - Explain how to use the program
 - Describe how the DSA works, including logic and examples
- Contributions will be evaluated based on personal documentation
- Deadline: 2026/5/18 23:59 (Submit via gradescope)

30-Second Introduction Video

- Only PMs need to submit the video
- Briefly introduce your project within 30 seconds
- Videos will be played at the event via projector
- Increases chances of earning extra credit
- No point deduction if not submitted
- Deadline 5/18 23:59 (Submit via NTU COOL)

Booth Exhibition

- Date: 2026/5/19 DSA Lecture time
- Location: NTU Second Activity Center(二活) Room 701-706
- Bring at least one device to demo your program
- Showcase your program at your booth
- Explain your project concisely to earn extra credit from the TAs
- Visit at least five other booths and vote for the coolest DSA

Grading Policy

Base Score: 3 (Basic) + 1 (Bonus) points

- 3 Basic points:
 - Evaluated by personal documentation
 - Demonstrating effort ensures full credit
- 1 Bonus point:
 - Earned based on presentation and program quality
- Group Peer Review Adjustment: Fill out a peer review form (rating yourself and your group mates from 1 to 5)
- Your score= **Basic + Bonus * [Your Received Peer Score Sum / (Total Members * 5)]**
- Not filling out the form will be considered as having zero contribution and will receive **0 point**
- "Total Members" count in the formula excludes anyone who didn't submit the form
- Peer review form will be released after the event

Hope you have fun!!!
